

Apache JMeter JSON Assertion Testing Report Case Study 3

1. Objective

The objective of this case study was to validate JSON responses returned by REST APIs using JSON Assertion in Apache JMeter.

The testing verifies whether the expected JSON elements and values are present in API responses.

2. Tool Used

- Apache JMeter 5.6.3
 - Java JDK
 - Windows Operating System
-

3. APIs Tested

API 1

GET /posts/1

URL:

<https://jsonplaceholder.typicode.com/posts/1>

Validation:

- userId = 1
- id = 1

JSON Path Used:

\$.userId
\$.id

API 2

GET /posts/1/comments

URL:

<https://jsonplaceholder.typicode.com/posts/1/comments>

Validation:

- postId = 1
- id = 1

JSON Path Used:

\$.posts[0].postId
\$.posts[0].comments[0].id

API 3

GET /comments?postId=1

URL:

<https://jsonplaceholder.typicode.com/comments?postId=1>

Validation:

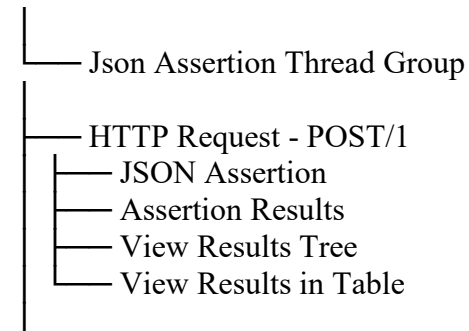
- postId = 1
- id = 1

JSON Path Used:

\$.comments[0].postId
\$.comments[0].id

4. Test Plan Structure

Test Plan



—	HTTP Request - /posts/1/comments
—	JSON Assertion
—	Assertion Results
—	View Results Tree
—	View Results in Table
—	HTTP Request - /comments?postId=1
—	JSON Assertion
—	Assertion Results
—	View Results Tree
—	View Results in Table

5. Test Execution

Thread Configuration

- Number of Users: 1
- Ramp-Up Time: 1 Second
- Loop Count: 1

Assertions Used

- JSON Assertion
 - Purpose:
 - Validate JSON structure.
 - Verify expected values.
 - Ensure API responses contain required fields.
-

6. HTML Dashboard Analysis

Test Information

Source File:

result3.jtl

Dashboard Generated Successfully.

APDEX Analysis

- APDEX Score: 1.000

- Interpretation:
- Excellent Performance
- Users received responses within acceptable limits.
- No performance degradation observed.

Request Summary

- PASS Requests: 100%
- FAIL Requests: 0%

Result:

All requests passed successfully.

7. Statistics

Overall Statistics

Metric	Value
Total Samples	3
Failed Requests	0
Error Percentage	0.00%
APDEX Score	1.000
Average Response Time	200.67 ms
Maximum Response Time	458 ms
Minimum Response Time	70 ms

Individual API Performance

GET /comments?postId=1

Link : <https://jsonplaceholder.typicode.com/posts/1>

Average Response Time: 70 ms

Error Rate: 0%

Status: Passed

GET /posts/1/comments

Link : <https://jsonplaceholder.typicode.com/posts/1/comments>

Average Response Time: 74 ms

Error Rate: 0%

Status: Passed

GET /posts/1

Link : <https://jsonplaceholder.typicode.com/comments?postId=1>

Average Response Time: 458 ms

Error Rate: 0%

Status: Passed

8. Assertion Verification

JSON Assertions successfully validated:

API 1

Verified:

- \$.userId = 1
- \$.id = 1

Result: Passed

API 2

Verified:

- \$[0].postId = 1
- \$[0].id = 1

Result: Passed

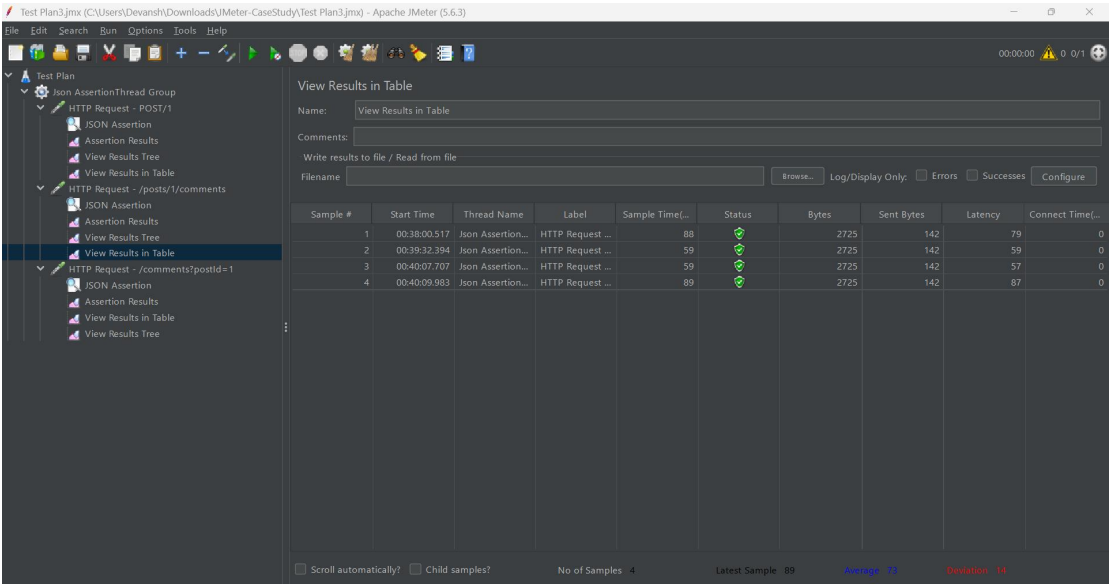
API 3

Verified:

- `$.postId = 1`
- `$.id = 1`

Result: Passed

9. Screen Shot



Test Plan3.jmx (C:\Users\Devansh\Downloads\JMeter-CaseStudy\Test Plan3.jmx) - Apache JMeter (5.6.3)

File Edit Search Run Options Tools Help

00:00:00 0 0/1

Test Plan

- Json AssertionThread Group
 - HTTP Request - POST/1
 - JSON Assertion
 - Assertion Results
 - View Results Tree
 - View Results in Table**
 - HTTP Request - /posts/1/comments
 - JSON Assertion
 - Assertion Results
 - View Results in Table
 - HTTP Request - /comments?postId=1
 - JSON Assertion
 - Assertion Results
 - View Results in Table

View Results in Table

Name: View Results in Table

Comments:

Write results to file / Read from file

Filename: Browse... Log/Display Only: ☐ Errors ☐ Successes

Sample #	Start Time	Thread Name	Label	Sample Time...	Status	Bytes	Sent Bytes	Latency	Connect Time...
1	00:35:34.490	Json Assertion...	HTTP Request	164		1514	133	163	108
2	00:35:35.880	Json Assertion...	HTTP Request	217		1514	133	217	145
3	00:35:36.439	Json Assertion...	HTTP Request	163		1514	133	162	108
4	00:38:00.285	Json Assertion...	HTTP Request ...	231		1514	133	231	152
5	00:39:32.219	Json Assertion...	HTTP Request ...	175		1514	133	175	117
6	00:40:07.537	Json Assertion...	HTTP Request ...	170		1514	133	169	111
7	00:40:09.725	Json Assertion...	HTTP Request ...	256		1514	133	256	169

☐ Scroll automatically? ☐ Child samples? No of Samples: 7 Latest Sample: 256 Average: 196 Deviation: 34

Test Plan3.jmx (C:\Users\Devansh\Downloads\JMeter-CaseStudy\Test Plan3.jmx) - Apache JMeter (5.6.3)

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 - HTTP Request - /comments?postId=1
 - JSON Assertion
 - Assertion Results
 - View Results in Table**
 - View Results Tree

View Results in Table

Name: View Results in Table

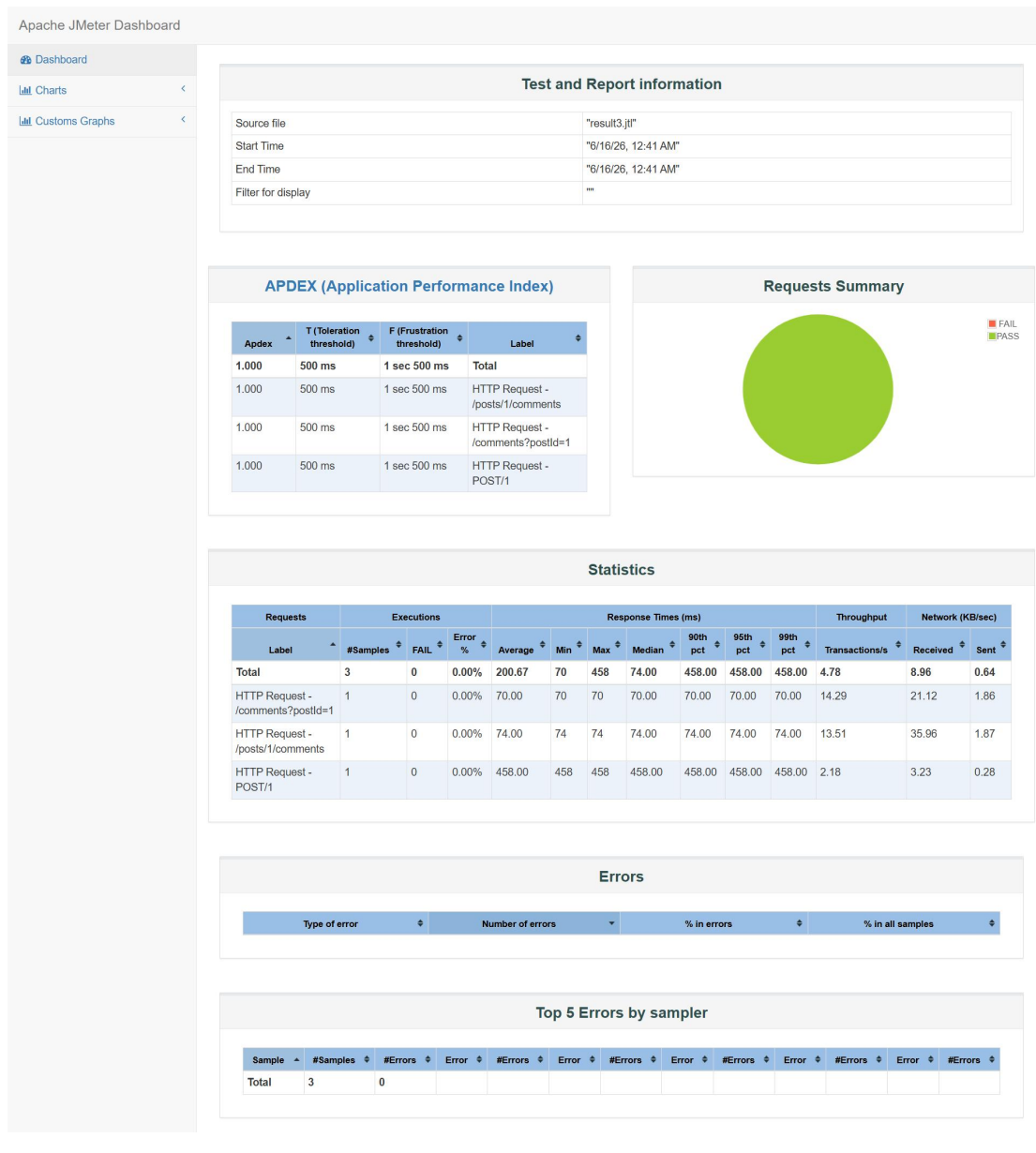
Comments:

Write results to file / Read from file

Filename: Browse... Log/Display Only: ☐ Errors ☐ Successes

Sample #	Start Time	Thread Name	Label	Sample Time...	Status	Bytes	Sent Bytes	Latency	Connect Time...
1	00:39:32.454	Json Assertion...	HTTP Request ...	55		1514	133	55	0
2	00:40:07.767	Json Assertion...	HTTP Request ...	56		1514	133	55	0
3	00:40:10.073	Json Assertion...	HTTP Request ...	79		1514	133	79	0

☐ Scroll automatically? ☐ Child samples? No of Samples: 3 Latest Sample: 79 Average: 60 Deviation: 11



10. Observations

1. All APIs returned HTTP Status Code 200.
2. JSON Assertions executed successfully.
3. Required JSON elements were found in responses.
4. No assertion failures occurred.
5. Error Percentage remained 0%.
6. HTML Dashboard confirms successful execution.
7. APDEX score indicates excellent application performance.

11. Conclusion

The JSON Assertion Case Study was successfully completed using Apache JMeter 5.6.3.

Three REST APIs were tested using JSON Assertions to validate response content. All assertions passed successfully, confirming that the APIs returned the expected JSON data. The generated HTML Dashboard reported 100% successful execution with zero failures and an APDEX score of 1.000.

Result: PASSED

Overall Status: SUCCESSFUL

Tool Used: Apache JMeter 5.6.3